

CO2 AT2

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COURSE CODE: CSA0918

Q) Design a generic method that finds the maximum element in a collection. Decide the required type bound and explain why an unbounded type parameter is insufficient. [Create]The challenge:

- The required type bound is **T extends Comparable<T>**. This ensures that the elements can be compared using `compareTo()`.
- An unbounded type parameter such as `<T>` is insufficient because Java does not know whether objects of type T can be compared.

Solution Strategy

- Start with the first element as the maximum.
- Traverse each element in the collection.
- Compare the current element with the maximum using `compareTo()`.
- If the current element is greater, update the maximum.
- Return the maximum element.

PROGRAM:

```
import java.util.*;
```

```
public class GenericMaximum {
```

```
    public static <T extends Comparable<T>> T findMaximum(Collection<T> collection) {
```

```
        Iterator<T> iterator = collection.iterator();
```

```
        T max = iterator.next();
```

```
        while (iterator.hasNext()) {
```

```
            T current = iterator.next();
```

```
            if (current.compareTo(max) > 0) {
```

```
                max = current;
```

```
            }
```

```
}

    return max;
}

public static void main(String[] args) {

    List<Integer> numbers = Arrays.asList(25, 78, 42, 91, 36);

    List<String> names = Arrays.asList("Althaf", "Zayan", "Raiyaan", "Arif");

    System.out.println("Maximum number: " + findMaximum(numbers));
    System.out.println("Maximum string: " + findMaximum(names));
}
}
```

OUTPUT:

Maximum number: 91

Maximum string: Zayan